———— **IDAMAN PHARMA MANUFACTURING** ————

CITREX GLUCOSAMINE 500MG CAPSULE

DESCRIPTION
Size 0, white body and scarlet cap, filled with white to off white powder.

ACTIVE: Glucosamine Sulfate Potassium Chloride 663.32 mg equivalent to Glucosamine Sulfate 500 mg.

SOURCE: The gelatin capsule source is from bovine. This product is derived from seafood (Chitin in the sea shrimp and crab shell)

ROUTE OF ADMINISTRATION
Oral

PHARMACODYNAMICS
Glucosamine is a natural substance found in chitin, mucoproteins, and mucopolysaccharides. It is involved in the manufacture of glycosaminoglycan, which forms cartilage tissue in the body; glucosamine is also present in tendons and ligaments. Glucosamine must be synthesised by the body but the ability to do this decline with age. Glucosamine and its salts have therefore been advocated in the treatment of rheumatic disorders including osteoarthritis. Glucosamine also acts to improve the viscosity of synovial fluid by increasing synovial fluid production, thereby providing lubricant activity.

PHARMACOKINETICS
Absorption:

After oral administration, bioavailability is low due to first-pass hepatic metabolism~26%. The gastrointestinal absorption is close to 90%.

Distribution:

Glucosamine is not protein-bound, but rather incorporates into plasma proteins (primarily globulins)

Volume of Distribution: 2.5 Liters

Metabolism: Liver, extensive. The first-pass effect in the liver in which more than 70% of glucosamine is metabolized.

Excretion: Renal excretion, 10% feces, 11% part of a dose of glucosamine sulfate is eliminated as carbon dioxide via expired air.

INDICATIONS
As adjuvant therapy for osteoarthritis.

RECOMMENDED DOSAGE
a) Light or Moderate Arthrosic Symptoms: 1 capsule twice daily for at least 6 weeks.
b) Severe Arthrosic Symptoms: Initial therapy: 1 capsule 3 times a day is recommended during a period of at least 8 weeks.
c) Follow-up Theraphy: Maintenance therapy should be followed for 3 – 4 months by administration of 1 capsule twice daily.
d) Long-term treatment: The treatment should be repeated every other 6 months or less as required.

CONTRAINDICATIONS
Contraindicated in patients hypersensitive to glucosamine and allergic to chitin in the sea shrimp and crab shell.

WARNINGS & PRECAUTIONS
Glucosamine may increase insulin resistance. Those with type 2 diabetes and those who are overweight and have problems with glucose tolerance should have their blood sugars carefully monitored if they consume glucosamine.

Patients with increased risk of cardiovascular diseases are advised to monitor the lipid level, as hypercholesterolemia has been diagnosed in a number of patients who were treated with glucosamine.

There is a report about exacerbated asthma symptoms that arose after the treatment with glucosamine was started (the symptoms resolved after the intake of glucosamine was after suspended). Asthma patients who start to take glucosamine should therefore be aware that the symptoms could exacerbate.

Glucosamine treats the underlying cause of osteoarthritis and the therapeutic effect can only be seen after 2 – 3 weeks. Therefore, it is advisable to take an analgesic/ anti-inflammatory drug if required during the first 2 -3 weeks of therapy with glucosamine.

Administration during the first three months of pregnancy must be avoided.

Safety and effectiveness have not been established in children. Therefore, children should avoid using glucosamine.

The administration in patients with severe hepatic or renal insufficiency should be made under medical supervision. The potassium level of glucosamine should be taken into account for patients with reduced renal function or controlled potassium diet.

As the active ingredient is derived from seafood, therefore should not be given to patients who are allergic to shellfish.

A doctor should be consulted in order to exclude the presence of other joint conditions/ diseases for which an alternative treatment should be considered.

INTERACTIONS WITH OTHER MEDICAMENTS
Effects on glucose metabolism and antidiabetic agents:
It has been hypothesized that glucosamine may impair insulin secretion through competitive inhibition of glucokinase in pancreatic beta cells and/or alteration of peripheral glucose uptake.

Glucosamine may increase insulin resistance and consequently affect glucose tolerance.

It may reduce antidiabetic agent effectiveness when used with these antidiabetic agents: Acarbose, Acetohexamide, Chlorpropamide, Glipizidede, Glyburide, Metformin, Miglitol, Pioglitazone, Repaglinide, Glimepiride, Tolbutamide and Troglitazone.

Glucosamine is likely safe in patients with well-controlled diabetes (HbA1c less than 6.5%) taking one or two oral antidiabetic medications or controlled by diet only. In patients with higher HbA1c levels or those taking insulin, monitor blood glucose level closely/ more frequently.

Reduced effectiveness when used with glucosamine: Doxorubicin, Etoposide and Teniposide.

An elevation of International Normalised Ratio (INR) serum and potentiation of anticoagulant effects were noted when glucosamine was administered to patient who is treated with warfarin. If concomitant therapy is necessary, the patient's INR should be more closely monitored.

Concomitant treatment of glucosamine with tetracycline can increase the absorption and the tetracycline serum concentration, but the clinical relevance of this interaction is probably limited.

Due to the limited documentation about possible interactions of glucosamine with other medicinal products, one should remain vigilant and note any changed reaction or concentration of medicinal products that are concomitantly used.

PREGNANCY & LACTATION

Available evidence is inconclusive or inadequate for use in pregnant or lactating mothers. Until more information is available, this product should only be used under medical supervision in pregnancy and lactating mothers if the potential benefit to the mother justifies the potential risk to the fetus.

Administration during the first 3 months of pregnancy must be avoided.

ADVERSE EFFECTS

a) Cardiovascular:
Peripheral oedema, tachycardia were reported in a few patients following larger clinical trials investigating oral administration in osteoarthritis. Causal relationship has not been established.

b) Central Nervous System:
Drowsiness, headache, insomnia and fatigue have been observed rarely during the therapy. (less than 1%)

c) Gastrointestinal:
Nausea, vomiting, diarrhoea, dyspepsia or epigastric pain, abdominal pain, constipation, heartburn and anorexia have been described rarely during oral therapy with glucosamine.

d) Skin:
Skin reaction such as erythema, and pruritus, have been reported with therapeutic administration of glucosamine.

e) Metabolism and nutrition disorder:
Hypercholesterolaemia.

SYMPTOMS AND TREATMENT OF OVERDOSAGE

Signs and symptoms of accidental or deliberate overdosing with glucosamine can include headache, dizziness, disorientation, arthralgia, nausea, vomiting, diarrhoea or constipation. In the event of overdosing, the treatment with glucosamine must be suspended and the standard supportive measures should be applied as required.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINE

No research has been performed with respect to the ability to drive and use machines. If undesirable effects such as headache, drowsiness or fatigue occur then you are advised not to drive or use machines.

INSTRUCTIONS FOR USE

Not applicable

STORAGE CONDITIONS

Store below 30ºC. Protect from light and moisture.

SHELF LIFE

2 years from manufacturing date.

PACK SIZES

Plastic container of 60 capsules & 90 capsules

REGISTRATION NUMBER

MAL20172942X

KEEP MEDICINES OUT OF REACH OF CHILDREN
JAUHI UBAT-UBATAN DARI KANAK-KANAK

For further information, please consult your doctor or your pharmacist.

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Lt-439.01 Product Registration Holder & Manufacturer
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